⑨ Option valuation — SM 7.1 to 7.3 in exam order

Asked: May 2026 Q6(b) 7m (Black-Scholes), Jan 2026 Q6(b) 6m and May 2026 MCQ CS-II (risk-neutral probability, real options).

Step 0 Pay-off at expiry (SM 7)

→ **Call** = Max (0, S − X) · **Put** = Max (0, X − S). Value can't be negative.

→ Every model answers: what is that future pay-off worth **today**? All use **risk-neutral valuation** (SM 7.3): option price is independent of risk preferences → use probabilities that make the share earn the **risk-free rate**, and discount at the **risk-free rate**.

Step 1 Building blocks

Symbol	Meaning	How
u	Up factor	Up price ÷ today's price
d	Down factor	Down price ÷ today's price
r	Risk-free growth factor for ONE step	e given → e^{rt} (Illus 2: $e^{0.02}$ = 1.0202) · steps of 6 months → $(1 + r)^{½}$ (Illus 1: 1.0247) · simple → 1 + r × t (TYK P22: 1.025)
p	Risk-neutral probability of UP	**p = (r − d) ÷ (u − d)**; down = 1 − p

Why p = (r − d)/(u − d): in a risk-neutral world the share grows at the risk-free rate → p × u + (1 − p) × d = r → solve for p.

SM Illus 2 writes the same in ₹: $(S_u - S)\,p + (S_d - S)(1 - p) = S\,(r - 1)$ → expected gain on share = risk-free interest on share.

Method A · Risk-neutral, one step (SM 7.2)

C = [p × C_u + (1 − p) × C_d] ÷ r — expected pay-off at p, discounted at risk-free.

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Illus 2 (Penchant): S 420 → 500 / 400 in 3 months, X 450, 8%, $e^{0.02}$ 1.0202 · 80p − 20(1 − p) = 420 × 0.0202 → **p 0.2848** · C_u 50, C_d 0 → C = 0.2848 × 50 ÷ 1.0202 = **₹ 13.96**. **SM example:** S0 → 60/40, 5% → p 0.625 → (10 × 0.625) ÷ 1.05 = **5.95**. **TYK P21 (EIL):** d 0.976, u 1.406, $e^{0.036}$ 1.037 → p = 0.061 ÷ 0.43 = **0.1418**.

Method B · Binomial tree, many steps (SM 7.1)

1. u, d, r per step → p.
2. Pay-offs at the **last** column.
3. Work **backwards** node by node with Method A's formula.

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Illus 1 (CAB Ltd, 2 steps of 6 months): 30 → 33/27 → 36.30/29.70/24.30, X 28, 5% p.a. · u 1.10, d 0.90, r = $(1.05)^{½}$ 1.0247 → **p 0.6235**, 1 − p 0.3765 · C_{uu} 8.30, C_{ud} 1.70, C_{dd} 0 · C_u = (0.6235 × 8.30 + 0.3765 × 1.70) ÷ 1.025 = **5.673** · C_d = 1.05995 ÷ 1.025 = **1.0341** · C_0 = (3.537 + 0.3893) ÷ 1.025 = **₹ 3.83**. **TYK P23:** S0, ±20%, 6% → p = (1.06 − 0.80) ÷ 0.40 = **0.65**; node B 13.49 → C_0 **8.272**.

→ SM: binomial can price **American** options — check early exercise at every node. Black-Scholes can't.

Method C · Riskless hedge (delta) portfolio (SM 7.1)

→ Buy Δ **shares** + write **1 call** so that the portfolio is worth the same whether price goes up or down → riskless → earns the risk-free rate → fixes the call price.

→ **Δ (hedge ratio) = $(C_u - C_d) \div (S_u - S_d)$** = change in option value ÷ change in share price = shares to hold per call written.

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TYK P22 (Mr Dayal): S 600 → 780 / 480 in 3 months, X 630, 10% (1.025) · C_u 150, C_d 0 → Δ = 150 ÷ 300 = **0.50** (buy 0.5 share per call) · value at expiry: up 0.5 × 780 − 150 = 240; down 0.5 × 480 = 240 → riskless · (300 − P) × 1.025 = 240 → **P ₹ 65.85** · expected return (real 60/40 used here only): expected value 150 × 0.60 = 90 → (90 − 65.85) ÷ 65.85 = **36.67%**.

Illus 1 "implied riskless hedge": ICAI Δ = (5 − 0) ÷ (33 − 27) = **5/6 share per call written** (pay-offs 33 − 28 = 5 and 0 at the one-period nodes).

Note (app-written): ICAI's own node values 5.673 and 1.0341 would give Δ = 4.639 ÷ 6 = 0.773. Write ICAI's 5/6.

A vs B vs C — the difference

	Risk-neutral (A)	Binomial tree (B)	Riskless hedge (C)
Idea	Expected pay-off at risk-neutral p, discounted at r_f	A repeated node by node backwards	Shares + written call made riskless → earns r_f
Formula	$p = (r - d)/(u - d)$; $C = [pC_u + (1 - p)C_d]/r$	Same at each node	$\Delta = (C_u - C_d)/(S_u - S_d)$
Periods	One	Many; handles American	Usually one
Real probabilities?	No	No	No for price; yes only for "expected return"
Gives	Option value	Option value	Option value + hedge composition

Same facts → A and C give the same price (two views of one no-arbitrage argument).

Black-Scholes (SM 7.3)

$$C = S \times N(d_1) - X \times e^{-rt} \times N(d_2)$$

$$d_1 = [\log_e(S/X) + (r + \sigma^2/2)\,t] \div \sigma\sqrt{t} \cdot d_2 = d_1 - \sigma\sqrt{t}$$ (= same with $r - \sigma^2/2$)

Piece	Meaning (SM)
$S \times N(d_1)$	Shares held — **$N(d_1)$ = hedge ratio** (continuous Δ)
$X\,e^{-rt}\,N(d_2)$	PV of strike × **$N(d_2)$ = probability spot at expiry > X** (exercise)
$\log_e(S/X)$	How far in the money today
$(r + \sigma^2/2)\,t$	Growth at risk-free + volatility adjustment, to expiry
$\div \sigma\sqrt{t}$	Turns it into number of standard deviations → read N( ) from table
N( ) from ICAI table	Table gives tail area: d positive → N = 1 − tail (N(0.50327) = 1 − 0.3072 = 0.6928); d negative → N = tail (N(−0.2976) = 0.3830); interpolate between rows

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Illus 3(i) (TIC Ltd): S 415, X 400, r 5%, σ 22%, t 0.25 · $\log_e$ 1.0375 = 0.03681 · (0.05 + 0.0242) × 0.25 = 0.01855 · σ√t 0.11 · d_1 0.5033, d_2 0.3933 · N 0.6928 / 0.6529 · C = 287.512 − 257.916 = **₹ 29.60** > market 25 → underpriced → **buy**. **(ii)** S 380 → d_1 −0.2976, d_2 −0.4076 → N 0.3830 / 0.3418 → **₹ 10.52**. **(iii) Put** = C − S + PV(X) = 10.52 − 380 + 395.03 = **₹ 25.55**. **(iv) Dividend** ₹ 10 in 2 months → PV = $10 \times e^{-0.05 \times 2/12}$ = 9.917 → S 398.083 → d_1 0.125, d_2 0.015 → **₹ 18.98**. **TYK P25:** 80, 75, σ 0.40, 6 months, 12% → d_1 0.5820, d_2 0.2992 → 57.57 − 43.61 = **₹ 13.96**.

→ **Assumptions (TYK T3):** European options · no transaction costs · short-term rate known and constant · no dividend · random walk · returns normally distributed · variance constant.

→ **Advantage:** speed. **Limitation:** can't price American options (values only at expiry).

Exam steps: pay-offs Max(0, S − X) → u, d, r (check compounding) → p → one step $[pC_u + (1-p)C_d]/r$ or tree backwards → "riskless / perfect hedge" → $\Delta = (C_u - C_d)/(S_u - S_d)$, buy Δ shares per call → σ, $\log_e$ (ln) or N( ) given → Black-Scholes d_1 → d_2 → N → C; put by parity; dividend → S − PV(div).